

**ADMINISTRATIVE INFORMATION****FMI No: 25299B****FMI Title: LightSpeed Ultra Regulatory Gantry Tilt Covers and Fan Airflow Upgrade**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☐ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☐ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☐ No Parts are required to complete this FMI.
☒ Batch with	25300, 25307
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked: 228117****FMI Instructions Part No.: 2324552-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25299B

FMI Title: LightSpeed Ultra Regulatory Gantry Tilt Covers and Fan Airflow Upgrade

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

Effectivity

CT LightSpeed Ultra Systems

Gantry Model Number: 2281177

Serial Numbers: 261214CN1 through 279518CN5.

Purpose

The purpose of this FMI is to install additional covers to the base of the gantry. When tilted and rotating the rotate frame of the gantry is exposed and presents a physical hazard. Safety and Regulatory Engineering has determined that the addition of these covers is necessary to protect the operator.

The gantry covers added by this FMI will need to be removed before performing a Detailed Calibration and then reinstalled at the end of the process. The Detailed Calibration software (see FMI25300) presents a warning message indicating the need for cover removal via a pop up window.

The fan bracket changes are necessary to increase airflow. Addition of the new covers restricts airflow and will cause the gantry to overheat. Moving the fans closer to the top cover exhaust vents results in the necessary volume of airflow to ensure that overheating does not occur.

NOTE: The fans must be installed when the covers are added.

Prerequisite FMI's

- 25300

Related FMI's

- None

Tools Required

Qty	Description	Part #
1	Electric Drill Motor	N/A
1	Center Punch	N/A
1	Vacum Cleaner	N/A
1	Tap Handle	N/A
1	5 mm ball end hex key	N/A
1	3/8 " Drive Torque Wrench	N/A
1	1/2 " Drive Ratchet	N/A
1	2 ounces Cutting Oil	N/A

People Power

Primary FE for 3.0 hours labor and 1 hour of travel.

2nd FE for 15 minutes labor and 1 hour of travel.

Furnished Materials (2325433)

Part #	Description	Qty
2324552-100	FMI Document	1
2325291	5.2 mm drill bit	1
2325292	6 mm tapered tap	1
2325668	1/2" drive 14 mm hex driver	1
46-170684P2	Loctite 271	1
46-170683P1	Loctite 222	1
2265650-2	Cover for HEMRC I/F Bd.	1
2258731-2	Gantry Cover Support Angle (vertical)	1
2258067-2	Gantry Cover Support Brace (diagonal)	1
2294827	Left Side Panel	1
2294827-2	Right Side Panel	1
46-328417P12	Screw 6mm X 12mm	8
46-328431P2	Washer, Plain 6.4mm X 18mm	8
46-328432P3	Washer, Lock 6.1mm X 11.8mm	8
2309247	Bracket, Fan Mounting RH-H3	1
2309248	Bracket, Fan Mounting LH-H3	1
2309912	Bracket, Fan Support RH	1
2309913	Bracket, Fan Support LH	1
2317478	Cover, Switch - Gantry Fan H3	2
2318266	Plate, Fan Support Brkt. RH H3	1
2318266-2	Plate, Fan Support Brkt. LH H3	1
2318340	Harness, Fan H3	2
46-208695P63	Heyco Part # 2140 SB-1000-12	1
46-208695P81	Heyco Part # 2240 SB-1500-21	1

Part #	Description	Qty
46-328417P33	Screw, Hex Socket 16mm X 35mm	1
46-328417P11	Screw, Hex Socket 6mm X 10mm	12
46-328417P3	4 mm Screw	3
46-208758P3	Tywraps (7.31" x .184")	10
46-208759P1	Screw-down Tywraps (7.81"x.184")	2
2103580	Screw, 4mm x 8mm	1
46-208747P2	Cable Tie Mount Adhesive Backed	1
2278989	Nylon Lanyard	4
46-208758P1	Tywraps (3.62" x .094 ")	10

Procedure

Right Side Panel Installation Procedure (2294827-2)

1. Remove gantry side, top and front covers.
2. Install display and control panel in service locations.
3. Turn off axial drive enable.
4. Carefully slide new side panel (2294827-2) into position from the front of the gantry.

If holes already exist at the Tap Template locations shown in Figure 2, skip to step 18. No second FE is required to assist with the FMI!

If no holes exist, continue to step 5.

5. Using a 5 mm ball-end hex tool, loosely fasten panel with two screws (46-328417P12) and washers (46-328431P2) at the front edge (see Figure 1).

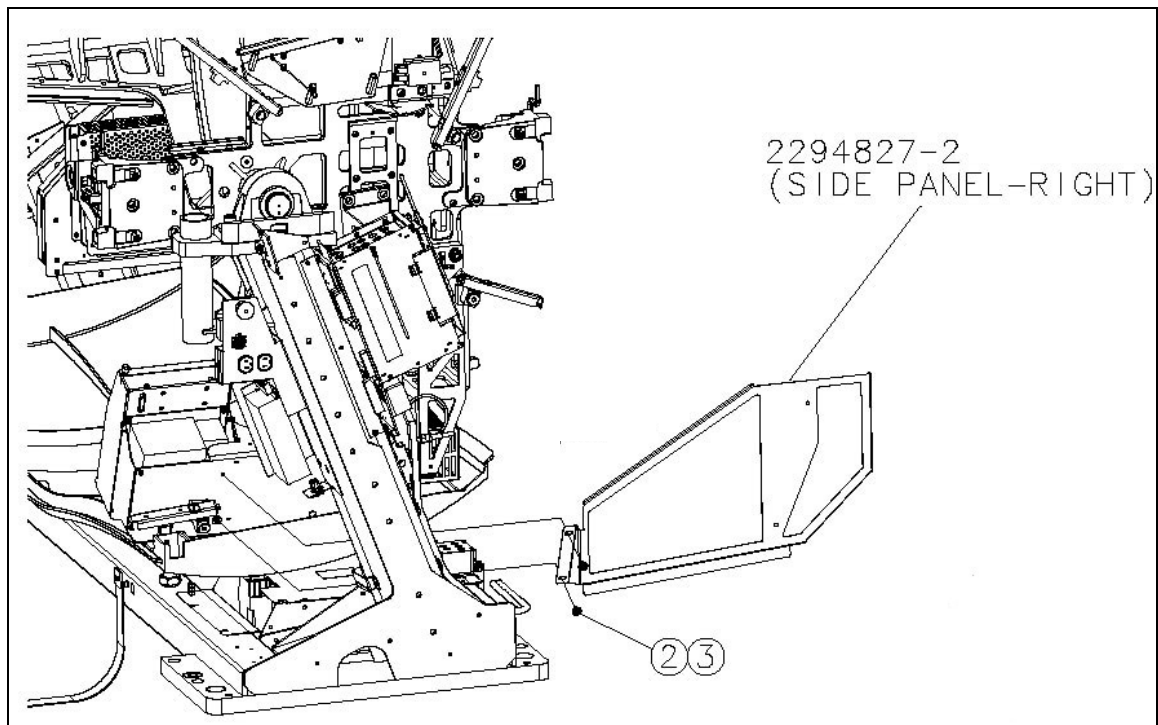


Figure 1. Right Side Panel Placement

6. Adjust panel to provide a clearance of about 1 cm as indicated in Figure 2, then tighten the two screws attached in step 4.

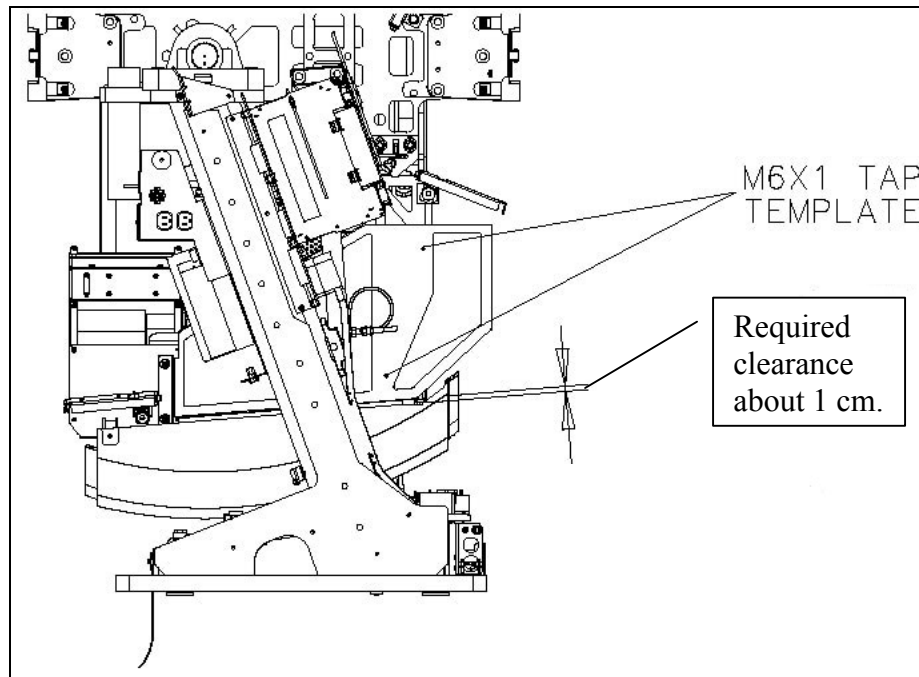


Figure 2. Right Side Panel Alignment

7. Tilt gantry forward to achieve a comfortable working position that provides access to the rear two holes on the cover.
8. Using the secured cover as a template, center punch the location of the holes for drilling.
9. Remove the cover and set it aside.
10. Place some protective cloths or towels on the gantry to catch metal shavings and prevent contamination of the gantry during the drilling and tapping process.
11. For each of the two holes, have the assisting FE plug in the vacuum cleaner and position its nozzle close to the drill site on the side you are drilling from.
12. Using the supplied 5.2mm drill bit, drill two holes for the additional mounting holes required for Right Cover (see Figure 2). Apply cutting oil to the drill site as necessary to ease the drilling process.
13. For each of the two holes, have the assisting FE position the vacuum cleaner nozzle close to the tap site on the side you are tapping from.
14. Using the supplied 6mm tap, tap the holes created in step 6. Apply cutting oil to the tap site as necessary to ease the tapping process.
15. Clean all surfaces of metal shavings and cutting fluid.

Second FE will not be required for the remainder of this FMI procedure.

16. Tilt gantry to 0 degrees.
17. Carefully slide new side panel (2294827-2) into position from the front of the gantry.
18. Loosely fasten two front screws (46-328417P12), two lock washers (46-328432P3), and two washers (46-328431P2) to the new holes at the front edge of the panel (see Figure 2).
19. Loosely fasten two rear screws (46-328417P12), two lock washers (46-328432P3), and washers (46-328431P2) to the new holes at the rear of the panel (see Figure 2).
20. Assure that the clearance indicated in Figure 2 is met before final tightening of the four screws.
21. Tighten screws.

Left Side Panel Installation Procedure (2294827)

1. Verify axial drive is turned off.
2. Inspect cover support brackets (Items 1 and 2, Figure 3) for threaded mounting holes. If holes exist in these brackets, skip to step 8. If holes do not exist, continue to step 3.
3. Remove the lower cover latches and cover support brace assembly for the front and rear cover (items 2, 4 and 5, Figure 3) from the tilting frame.

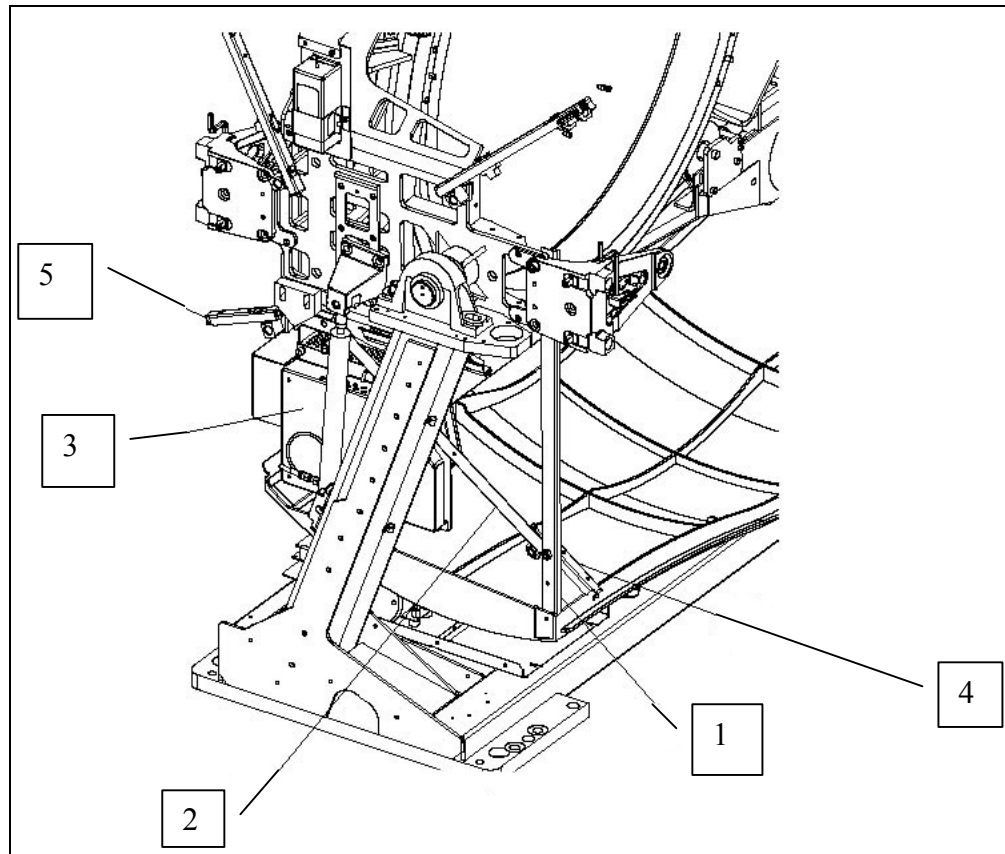


Figure 3. Left Side Panel Preparation

4. Remove the lower cover latches (items 4 and 5, Figure 3) from the cover support brace (item 2, Figure 3).
5. Apply Loctite 271 to each fastener and install lower cover latches (items 4 and 5, Figure 3) onto new cover support brace (2258067-2). Be sure to maintain same orientation of cover support brace to latches.
6. Remove gantry cover support angle (item 1, Figure 3) and replace with new angle supplied in FMI kit (2258731-2). Per Figure 4, apply Loctite 271 to each fastener.
7. Install the lower cover mounts and new support brace assembly (items 2, 4 and 5, Figure 3). Per Figure 4, apply Loctite 271 to mounting screws.

8. Tilt gantry to 0 degrees.
9. Shut off and verify gantry 550 VDC and 120 VAC power.
10. Replace Cover for HEMRC I/F Board (item 3, Figure 3) with cover (2265650-2) supplied in FMI kit.

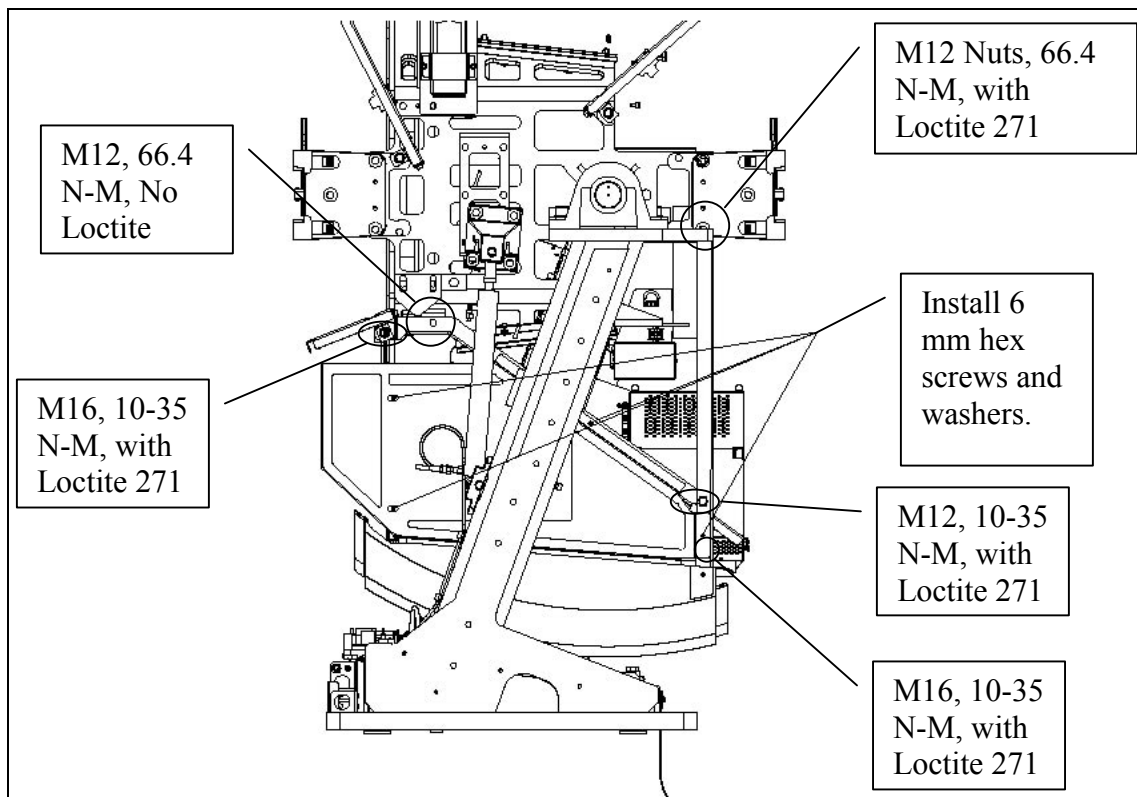


Figure 4. Left Side Panel Installation

11. Install Left Side Panel (2294827), using four screws (46-328417P11), four lock washers (46-328432P3), and four washers (46-328431P2) supplied in FMI kit (refer to Figure 4).
12. Torque all fastening nuts and screws as indicated in Figure 4.

Note: 1 Newton-Meter = 0.73692 Foot-Pounds

Right Side Fan Assembly Modification (2315618)

1. Turn off and verify Axial Drive Enable, 550 VDC and Gantry 120 VAC power.
2. Remove rear cover.
3. Disconnect the fuse wiring, the tape switch connector, and the port cable connection at the comm box (refer to Figure 5).

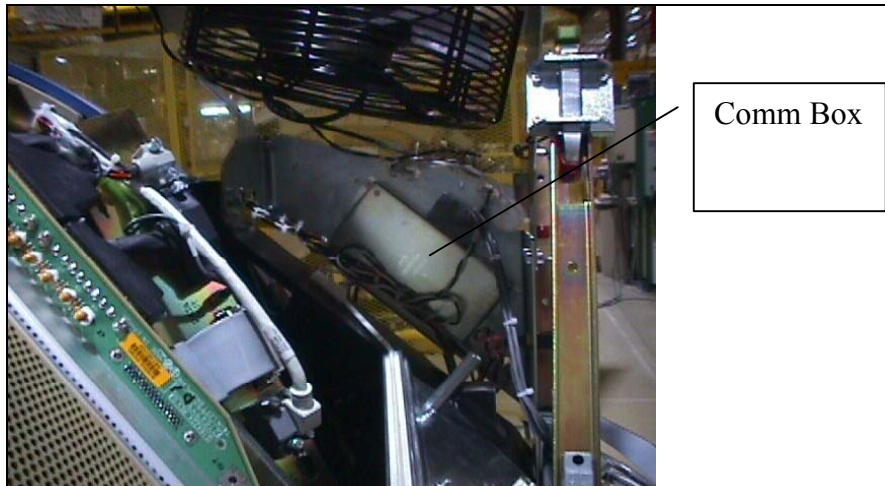


Figure 5. Comm Box

4. Remove the fuse assembly (item 8, Figure 6) and install on new assembly.
5. Remove the AC power supply in the back (item 4, Figure 6) that is fastened through the front. It can be left attached to the harness and hanging next to the assembly.
6. Pull fuse and contact switch wiring harness through the fan bracket assembly grommet and remove the AC filter in the back (item 18, Figure 6).

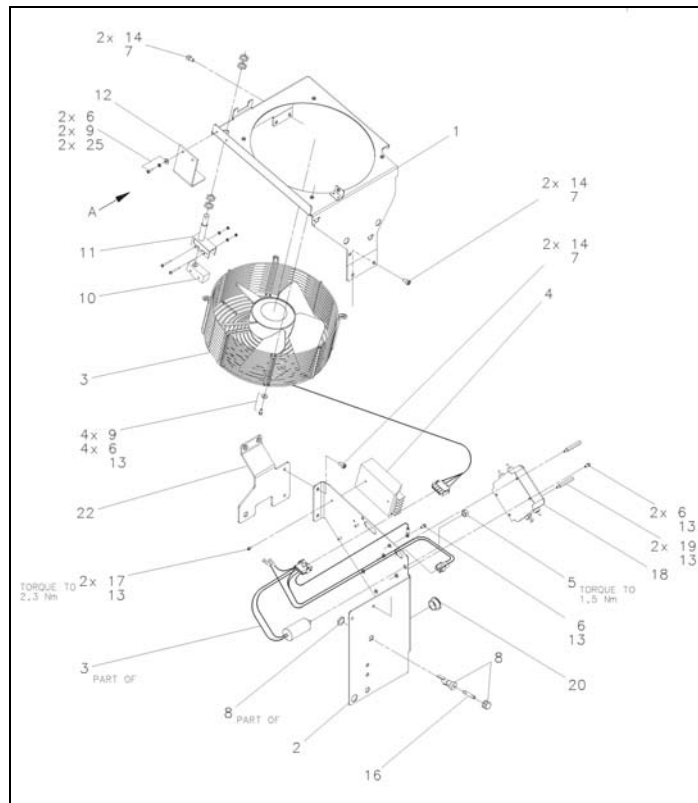


Figure 6. Complete Right Hand Fan Assembly

7. Disconnect the AC power lines from the fan harness.
8. Using the provided 14 mm 1/2" drive hex driver for the lower left bolt and your 10 mm hex wrench, remove entire fan assembly from the gantry.
9. Remove the Lexan cover (item 12, Figure 6), covering the limit switch, from the assembly.
10. Disconnect the limit switch (items 10 and 11, Figure 6) from the old cable harness. Note harness position on switch (Normally Open (NO) and Common Terminals).
11. Remove the limit switch and install on new fan mounting bracket assembly.
12. Adjust limit switch position per Figure 7.

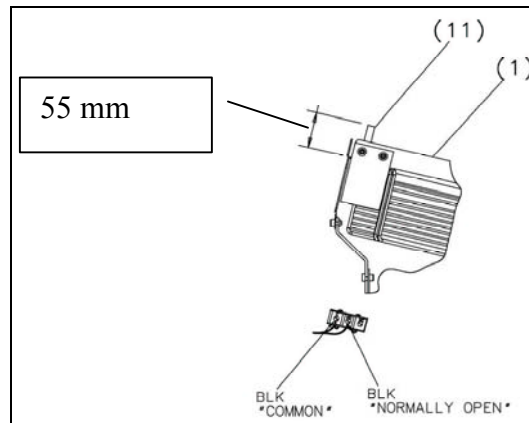


Figure 7. Limit Switch Height Adjustment

13. Install new Lexan cover (2317478) (item 12, Figure 6) on new assembly.
14. Remove fan (item 3, Figure 6), apply Loctite 222 to each screw and install fan on new assembly (item 1, Figure 6) (2309247).
15. Remove comm box (Figure 5) and install on new assembly.
16. Using the provided 14 mm 1/2" drive hex driver for the lower left bolt and your 10 mm hex wrench, install new assembly (Figure 8) onto gantry. Apply Loctite 271 to each bolt.

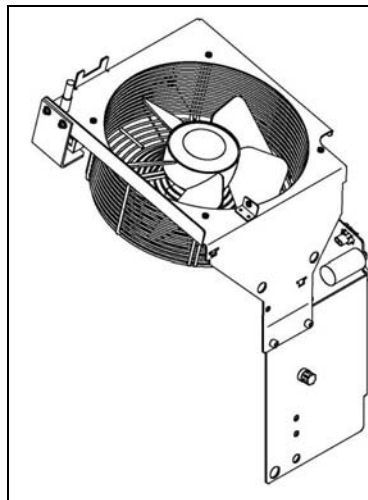


Figure 8. Complete Right Hand Fan Assembly

17. Per Figure 9, route new fan harness.
18. After applying Loctite 222 to each fastener, reattach the AC filter and AC power supply to the back of the bracket.

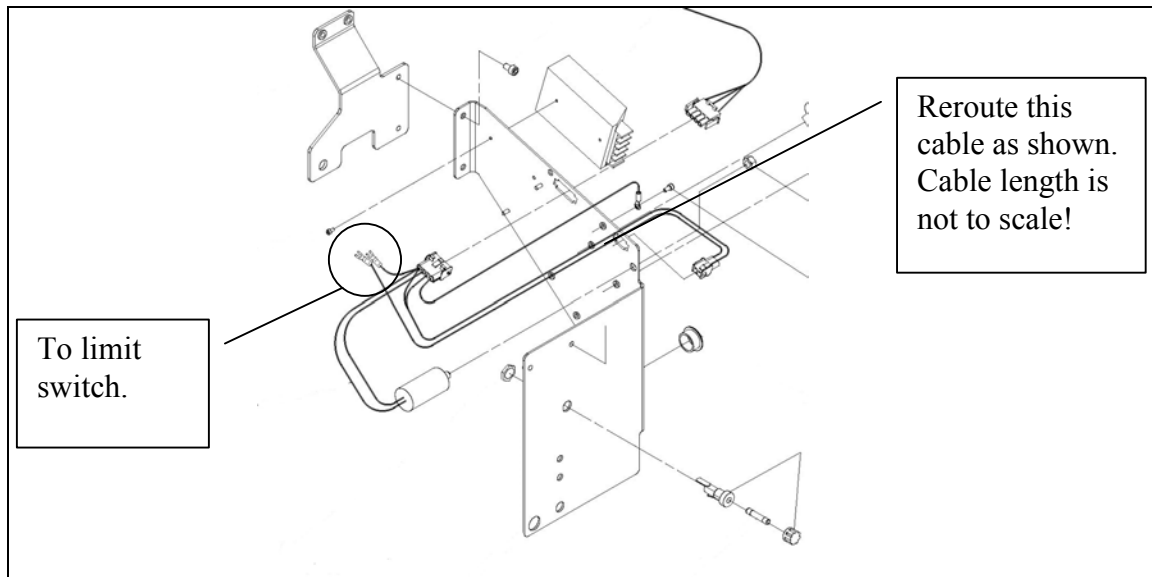


Figure 9. New Fan Harness Routing

19. Refer to figures 10, 11 and 12. Disconnect four wires on supply side of AC Filter.
20. Reroute 120V harness through fan bracket assembly grommet, tywrap as indicated and reconnect wires to AC Filter. Utilize tywraps as needed to ensure cables are dressed and routed as shown and that no harness is able to rub against sheet metal edges or corners.

NOTE: Verify the harnesses are not rubbing up against sheet metal edges or corners that may cut the wires when vibrating. You may have to cut tywraps on 120V harness in order to obtain sufficient slack to avoid contacting the edges of the fan bracket assembly.

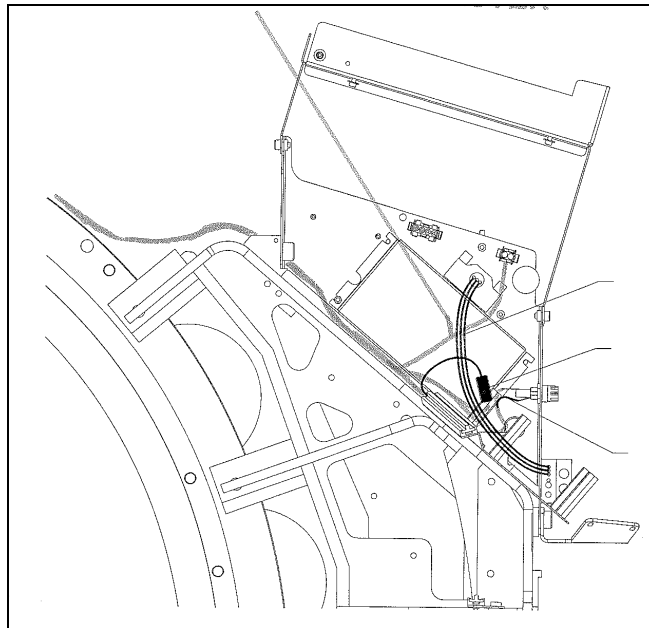


Figure 10. 120 VAC/Encoder Cable Routing – Gantry Right Hand Fan, Front

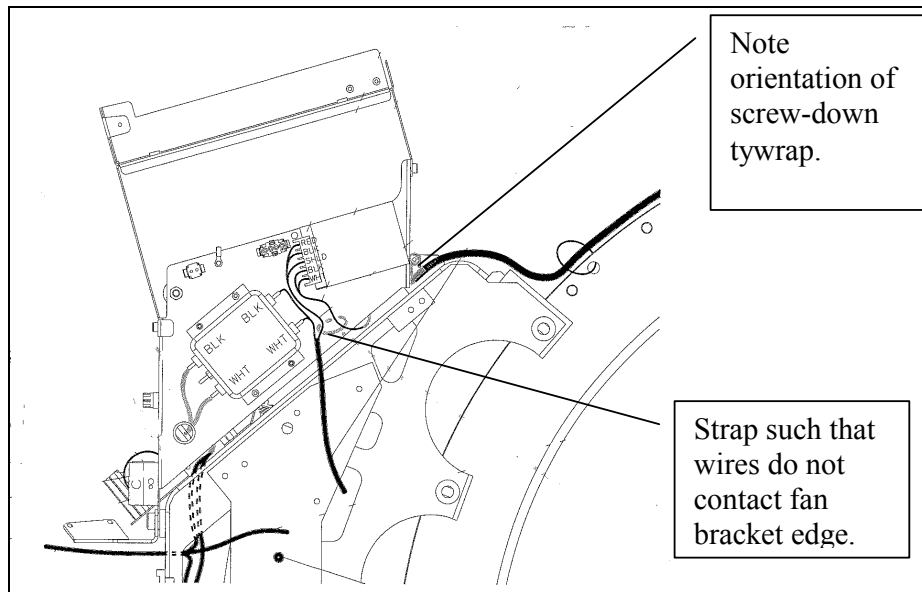


Figure 11. 120 VAC/Encoder Cable Routing – Gantry Right Hand Fan, Rear



Figure 12. Completed Cable Rerouting – RH Fan

21. Reconnect the fuse wiring, tape switch connector, and comm box port cable.
22. Reconnect AC to the fan harness.

Left Side Fan Assembly Modification (2315619)

1. Disconnect the encoder serial connectors and unfasten them from the bracket assembly.
2. Disconnect the AC from the fan harness.
3. Disconnect the Thermostat from the 120V harness (item 12, Figure 13) and remove from fan bracket assembly.
4. Cut tywraps holding 120 VAC harness to fan bracket assembly.
5. Remove Fan assembly from the gantry.
6. Remove the Lexan switch cover (item 11, Figure 13) from the assembly.
7. Disconnect the limit switch (items 9 and 10, Figure 13) from the old cable harness. Note harness position on switch (Normally Open (NO) and Common Terminals).

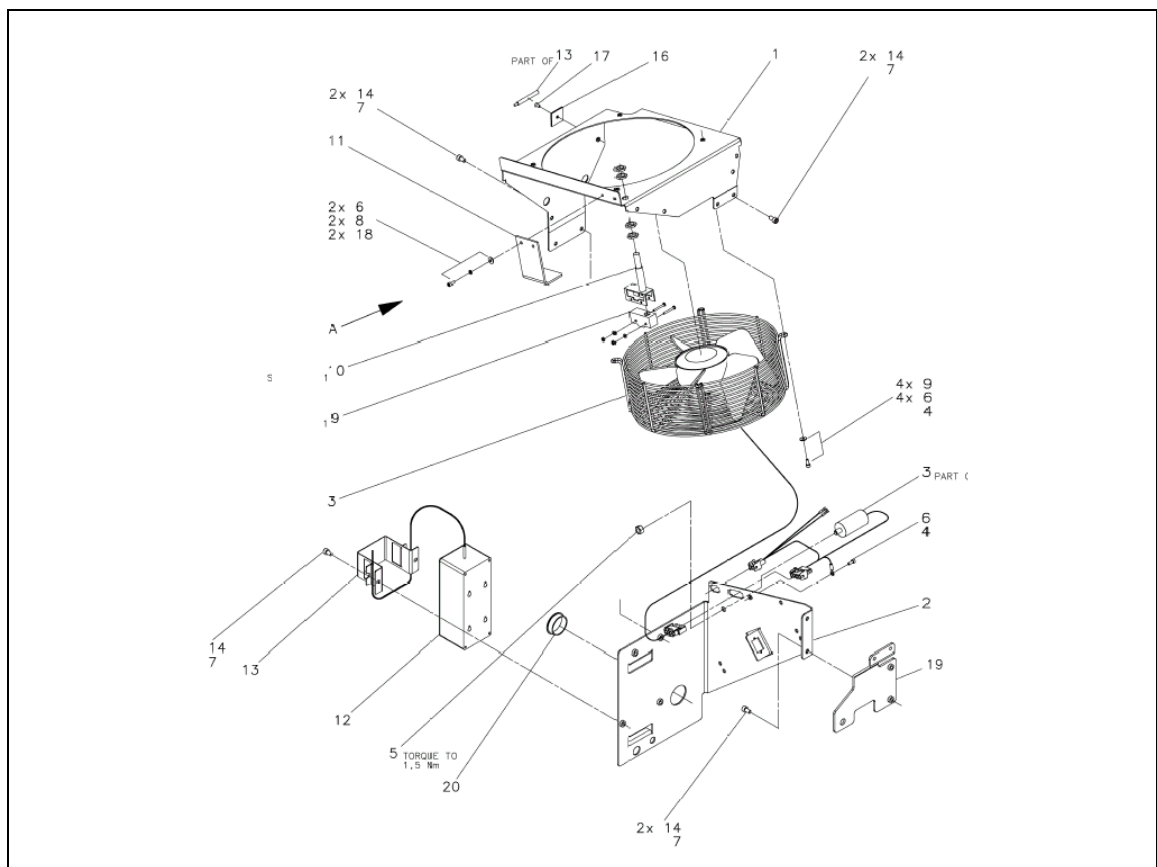


Figure 13. Complete Left Hand Fan Assembly

8. Remove the limit switch (items 9 and 10, Figure 13) and install on new assembly.
9. Per Figure 14, adjust limit switch height to 55 mm.

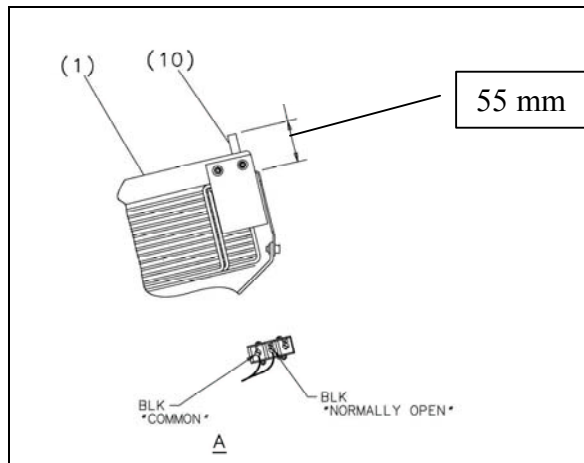


Figure 14. Limit Switch Height Adjustment

10. Install new Lexan switch cover (2317478) (item 11, Figure 13) on new assembly.
11. Remove fan (item 3, Figure 13), apply Loctite 222 to each screw and install fan on new assembly.
12. Install new assembly (Figure 15) on the gantry.
13. Using the provided 14 mm 1/2" drive hex driver for the new lower right bolt, add provided M16x35 (46-328417P33) bolt to fasten fan bracket assembly to tilt frame. Apply Loctite 271 to each bolt.
14. After applying Loctite 222 to each fastener, install thermostat (items 12 and 13, Figure 13) on new assembly.

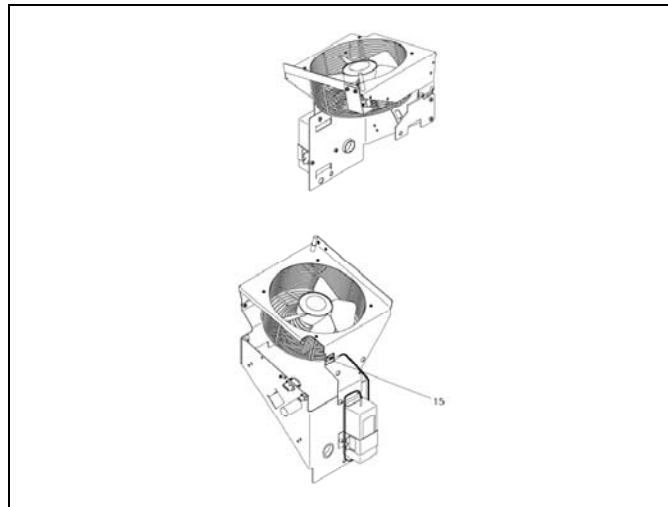


Figure 15. Completed Left Hand Gantry Fan Assembly

15. Attach thermostat sensor to tywrap mount (item 15, Figure 15).
16. Reattach the encoder serial cable.
17. Refer to Figure 16. Reroute 120V harness and encoder harness. This may require disconnecting some wires. Tywrap as indicated.
18. Refer to Figure 16 and 17. Utilize tywraps as needed to ensure cables are dressed and routed as shown and that no harness is able to rub against sheet metal edges or corners.

NOTE: Verify the harnesses are not running up against sheet metal edges or corners that may cut the wires when vibrating.

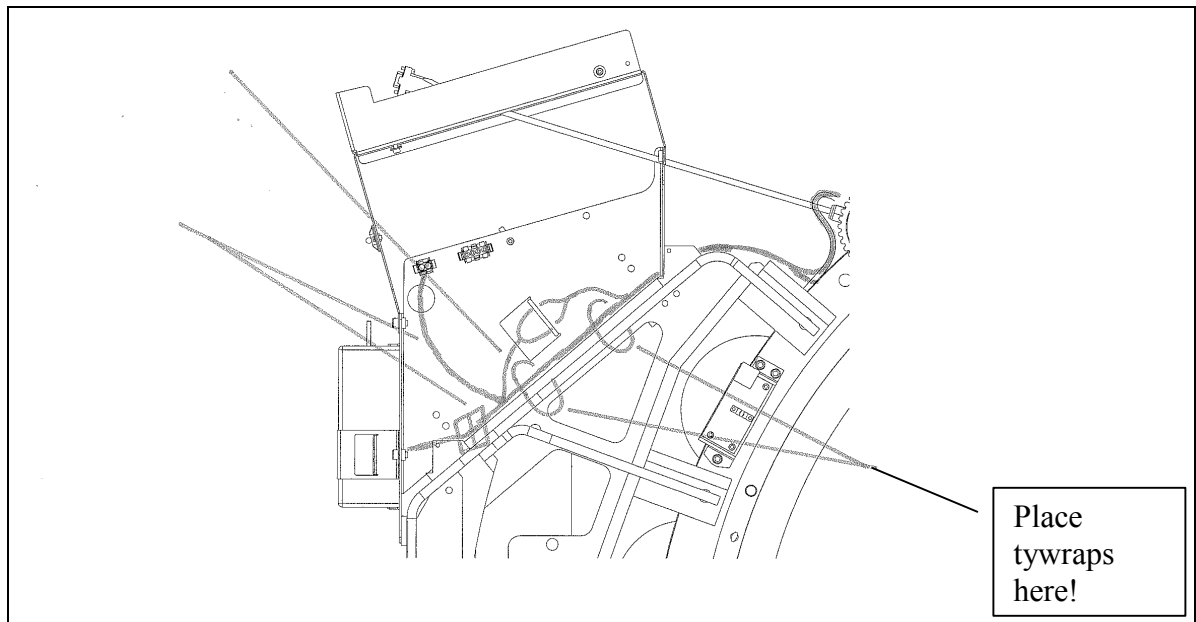


Figure 16. Cable Routing - Gantry Left Hand Fan, Front

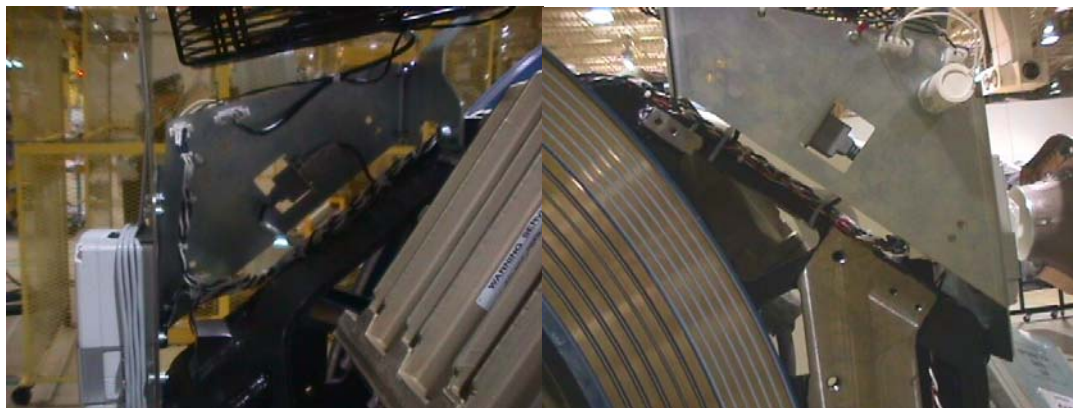


Figure 17. Completed Cable Rerouting – LH Fan

19. Reconnect the thermostat (item 12, Figure 13) to the 120V harness.
20. Reconnect AC to the fan harness.
21. Install rear gantry cover.
22. Place top covers onto gantry and verify limit switches engage. Referring to Figure 14, adjust as need to obtain positive actuation of the switches.

Note: The new limit switch location does not hit the pad on the top cover – this is expected.

23. Restore the 120V power.

24. Verify that the fans are working by setting the thermostat very low thus triggering fan operation.
25. Set the thermostat to normal operating settings of "26" and "2" after verification.

Finish Up

1. Turn all power back on
2. Install all covers.
3. Inspect all gantry cover attachments for proper fit and latching.
4. Reset system.
5. Tilt gantry to 0 degrees and verify that the covers do not scrap or interfere with stationary parts of the gantry.
6. Tilt gantry to +30 and -30 degrees and verify that the covers do not scrape or interfere with stationary parts of the gantry.
7. Verify scanner is ready for customer use.
8. Recycle left over materials.